Haoran (Jensen) Yuan

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EDUCATION

Doctor of Philosophy in Computer Engineering — Aug. 2026 – May 2028 (Expected)
University of Illinois Urbana-Champaign | Advisor: Prof. Ismini Lourentzou — Champaign, IL

Master of Science in Computer Engineering — Sep. 2024 – May 2026
University of Illinois Urbana-Champaign — Champaign, IL

Bachelor of Science in Electrical Engineering, Minor in Computer Science — Sep. 2020 – Jun. 2024
University of Illinois Urbana-Champaign — Champaign, IL

Bachelor of Engineering in Computer Engineering — Sep. 2020 – Jun. 2024
Zhejiang University — Hangzhou, China

RESEARCH INTERESTS

Efficient AI, with a focus on making **embodied AI generalizable, scalable, and efficient**: world-action models, visuo-tactile learning, lossless speculative decoding, agentic robotics, and reinforcement learning for world models.

PUBLICATIONS

** equal contribution † project lead*

[1] **DexTacWAM: A Visuo-Tactile World-Action Model for Dexterous Manipulation**
H. Yuan†, Z. Wang, B. Shao, H. Lu, T. Darrell, I. Lourentzou, W. Zhan. *arXiv:2609.24976*, 2026.
[project] [paper] [code]

[2] **VTAM: Video-Tactile-Action Models for Complex Physical Interaction Beyond VLAs**
H. Yuan*†, W. Yi*, Z. Zhang*, W. Chen*, Y. Mo, J. Yin, X. Li, X. Zeng, C. Wen, C. Lu, K. Driggs-Campbell, I. Lourentzou. *arXiv:2603.23481*, 2026.
[project] [paper]

[3] **JetSpec++: Foresight Parallel Drafting with Self Refinements**
Z. Feng*, L. Hu*, **H. Yuan**, Y.-Y. Qian, J. Yang, P. Zhao, Y. Zhu, T. Rosing, H. Zhang. *arXiv:2610.19792*, 2026.
[paper] [code]

[4] **JetSpec: Breaking the Scaling Ceiling of Speculative Decoding with Parallel Tree Drafting**
L. Hu, Z. Feng, Y. Wu, **H. Yuan**, Y. Zhao, Y.-Y. Qian, B. Wang, P. Zhao, D. Jiang, Y. Zhu, T. Rosing, H. Zhang. *arXiv:2606.18394*, 2026.
[project] [paper] [code]

RESEARCH EXPERIENCE

DexTacWAM: Visuo-Tactile World-Action Model (Project Lead, BAIR @ UC Berkeley) | *World Model, Tactile*

- Led a visuo-tactile **World-Action Model** that adapts a pretrained video DiT to 10 fingertip tactile sensors on a 22-DoF bimanual robot via **modality-split flow matching**, modeling contact that cameras miss under occlusion.
- Designed a finger- and pose-aware **tactile compressor** fusing 10 fingertip latents into 2 hand views, retaining **89.4%** of contact recall while cutting DiT tokens by **3.67**× and training time by **2.26**×.
- Achieved **70.6%** avg. success on 6 real-robot tasks vs. **38.0%** for the best baseline (RDP) with only ∼100 demos per task; featured by BAIR.

VTAM: Video–Tactile Action Model (Project Lead, PLAN Lab @ UIUC) | *PyTorch, World Model, Multimodal Learning*

- Led end-to-end development (2k+ real-world trajectories), adapting a pretrained video transformer to touch **without tactile pretraining**; reached **90% success** on chip pick-and-place, plate wiping and cucumber peeling.
- Designed a gated cross-attention mechanism to suppress cross-modal contamination between tactile and visual tokens, reducing training loss spikes by **42%** and speeding up convergence by **1.7**×.
- Added a force-aware diffusion regularizer to balance tactile–vision attention: **+80%** success over vision-only.

JetSpec++: Foresight Parallel Drafting with Self Refinements (Core Contributor, Hao AI Lab @ UCSD)

- Developed a **causal correction for parallel drafting** that conditions each drafted token on its **full branch history** rather than the preceding token alone, pre-drafting the next block during verification.
- At matched parameter count, **outperformed DeepSeek's DSpark**, with the margin widening at draft depths beyond those used in training; raised average acceptance length to **8.94** (vs. 6.89).
- Achieved up to **7.45**× lossless speedup over AR and **1.54**× over DSpark on Qwen3-8B, reaching **1,361 tok/s** for a single request on GB200.

JetSpec: Parallel Tree Drafting for Speculative Decoding (Core Contributor, Hao AI Lab @ UCSD)

- Co-developed a **causal parallel draft head** that emits a complete scored speculative tree in **one forward pass** while preserving the target model's autoregressive factorization along every branch.
- Integrated with a standalone inference engine using paged FlashAttention **tree-attention kernels** in Triton and CuTe DSL, plus vLLM serving support.
- Achieved up to **9.64**× lossless end-to-end speedup over AR on Qwen3-8B (MATH-500, H100), beating DDTree (8.78×) and EAGLE-3.

WORK EXPERIENCE

Snowflake | Software Engineering Intern — Menlo Park, CA | **May 2025 – Aug. 2025**

- Designed and implemented a **predictive autoscaler** for Snowflake GS clusters, combining Gradient Boosting, ARIMA and XGBoost models trained on historical telemetry from the Snowhouse database.
- Achieved **14.3% MAPE** and **92.5% coverage** at the 95% confidence interval for **24-hour** CPU forecasts using under one month of history, with end-to-end training and prediction in <5 minutes per deployment.
- Deployed on the internal background job system to adjust instance provisioning, increasing average **CPU utilization by >20%** and reducing overprovisioning in simulated datacenter environments.

TECHNICAL SKILLS

Languages: Python, C/C++, Rust, Java, SQL, SystemVerilog.
Frameworks: PyTorch, CUDA, Triton, SGLang, vLLM, Isaac Gym, Isaac Sim, MuJoCo, Flask, SQLAlchemy, OpenCV.
Databases & Tools: MySQL, PostgreSQL, Redis, Docker, AWS, GCP, Git, Linux, Shell.